Certified structural properties of the curve catalogue

Fixed-model classification and independent replay

11 October 2026

Native arithmetic and PARI agree on all 814 exact models. We certify 360 ordinary 2-volcano levels and retain bounds for 454. Among the 512 independent 192–252-bit sources, 219 are certified at the 2-surface and 65 have full rational 4-torsion. Original model identities and family/class labels are retained.

1 Inventory and evidence

The 277 $p = 7$ route/conversion/control models are postselected; 25 higher-degree models are constructed controls. The 512 independent large sources retain their original model law, including full rational 2-torsion by construction. All retained models are ordinary. Counts describe these inventories, with no uniform-class density or sampling-error interpretation. Field sizes are 7^6 (17 bits) and 192–252 bits, in degrees 6, 10 and 14. Selected subgroups and generators remain unspecified.

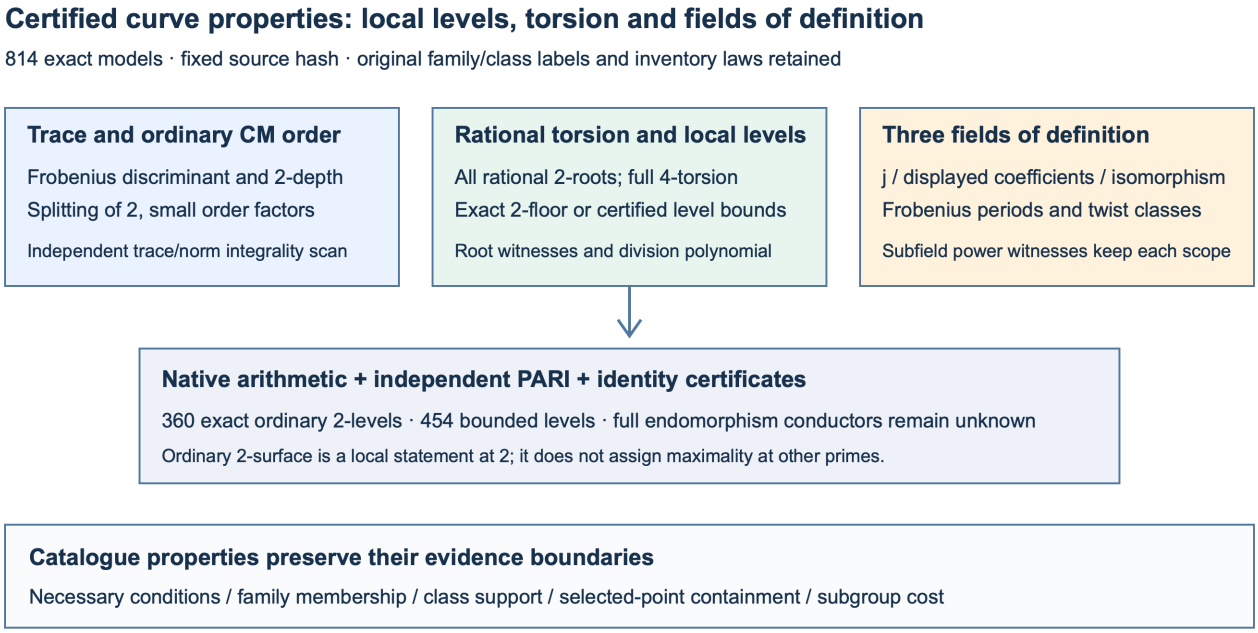


Figure 1: Independent structural properties feed the existing catalogue without promoting a necessary filter to family membership.

Inventory	Models	Full 2	Full 4	Exact level	Bounded	Proper iso
p7 replay/control	277	200	52	116	161	3
Higher-degree controls	25	0	0	25	0	0
Independent large sources	512	512	65	219	293	0

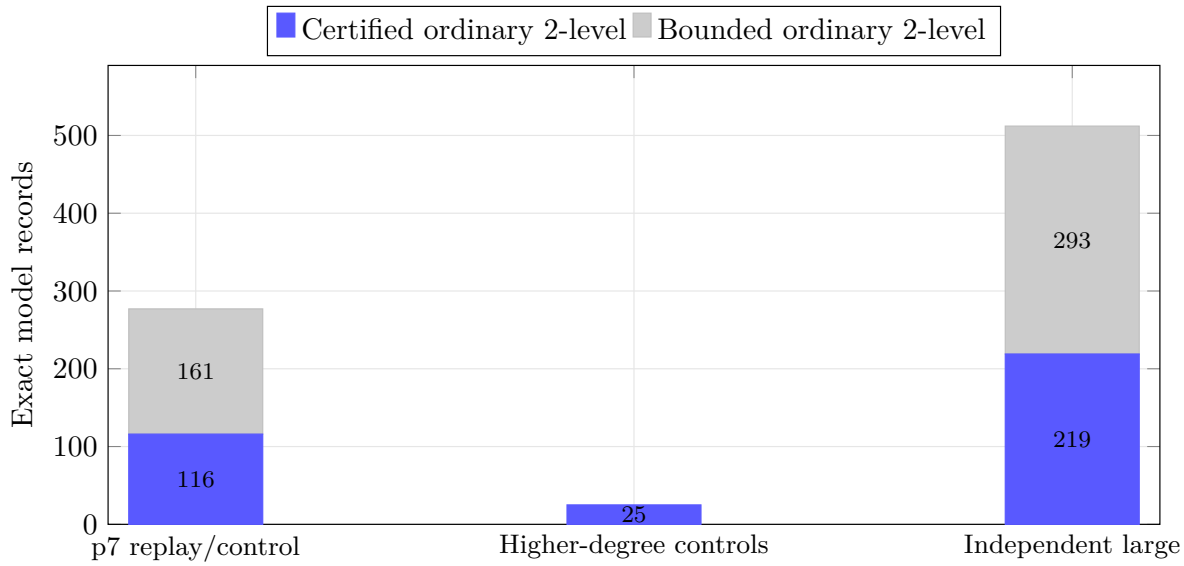


Figure 2: Exact and bounded ordinary 2-levels in each frozen model cohort. Units are model records; cohort input laws differ.

2 Local conductor data without odd factorization

For an ordinary trace write $\Delta = t^2 - 4Q = 2^s u < 0$ with u odd. Fundamental-discriminant classification gives

$$v_2(D_K) = \begin{cases} 3 & s \text{ odd}, \\ 0 & s \text{ even and } u \equiv 1 \pmod 4, \\ 2 & s \text{ even and } u \equiv 3 \pmod 4. \end{cases} \qquad v_2(f_\pi) = \frac{s - v_2(D_K)}{2}.$$

Odd squares are one modulo eight, so the odd fundamental-discriminant residue determines split/inert behavior of two; a positive two-valuation is ramified. Full D_K and the odd conductor remain uncomputed.

PARI independently searches integral $(\pi - a)/2^k$, whose trace and norm must both be integers:

$$\frac{t - 2a}{2^k}, \qquad \frac{Q - at + a^2}{2^{2k}}.$$

For even t there are two possible a residues modulo 2^k; odd t permits no positive k. In a maximal-order basis $\pi = A + f_\pi \omega$, the largest admissible denominator is exactly $v_2(f_\pi)$. This method checks every ordinary model, with a full-discriminant cross-check on the small discriminants.

3 Torsion certifies ordinary 2-levels

Proposition. Let $E/\mathbb{F}_Q$ be ordinary in odd characteristic and let $s = v_2(f_\pi)$. Exactly one rational nonidentity 2-point implies $v_2(f_E) = s$, the ordinary 2-volcano floor. Full rational 2-torsion implies $v_2(f_E) \le s - 1$; full rational 4-torsion implies $v_2(f_E) \le s - 2$. A collapsed bound certifies the local level.

Proof. In a basis $1, \omega$ of $\mathrm{End}(E)$, write $\pi = a + b\omega$. The order index is $f_\pi = f_E|b|$. Full rational $E[2]$ is equivalent to $\pi - 1 \in 2\,\mathrm{End}(E)$: an endomorphism killing $E[2]$ factors through the separable map $[2]$. If b is even, the odd norm Q forces a odd, giving this divisibility. Absence of full rational 2-torsion therefore forces b odd and the floor equality. Full rational $E[4]$ similarly implies divisibility by four and the second bound. □

The native checker validates all rational 2-roots and uses their differences and the square class of -1 to certify full 4-torsion. PARI independently uses the sextic factor of the degree-four division polynomial and the square classes of its y coordinates. Whole endomorphism conductors remain unknown; a 2-surface statement does not assert maximality at other primes.

4 Three fields of definition

The least Frobenius period of j determines its degree; the joint period of a, b determines the displayed model's degree. A K-isomorphic model can have a different minimum degree.

For generic j, set $a_0 = 3j/(1728 - j)$ and $b_0 = 2j/(1728 - j)$. The twist parameter is $D = 3b/(2a)$, with $a = D^2 a_0$ and $b = D^3 b_0$. At $j = 0$ use b modulo sixth powers, and at $j = 1728$ use a modulo fourth powers.

For a candidate $\mathbb{F}_{p^d}$, put $M = p^n - 1$, $m = p^d - 1$ and $h = \gcd(\text{twist order}, M/m)$. The image subgroup $\mathbb{F}_{p^d}^*(K^*)^{\text{twist order}}$ is generated by the h-th power of a cyclic generator. Thus the twist parameter lies in it exactly when its M/h power is one. Each row retains all candidate power witnesses, including rejected smaller fields.

Four $p = 7$ records have j degree three. Three have a K-isomorphic model over $\mathbb{F}_{7^3}$, and one requires $\mathbb{F}_{7^6}$. Only one already displays coefficients over $\mathbb{F}_{7^3}$:

Exact model slug	j	Coefficients	Isomorphic model
`icv1-fp3k6-tm110-5f712c89`	3	6	3
`icv1-fp3k6-t202-39e02f24`	3	6	6

These are model-definition properties. Selected-point and subgroup containment remain unspecified.

5 Certificates and verification

Thirteen portable identity certificates cover the cubic discriminant, cleared normalized-j and twist coefficients, translated Frobenius norm and eight unit-group index degrees. Each passes 64 saved and 128 fresh evaluations; native expansion additionally checks exact integer coefficients. IDs and expression trees are in `identity_certificates.json`; the formal-proof pointer remains null.

Native arithmetic and PARI agree on all 814 property rows. Audit run 2 completes under its 300-second cap with empty stderr. The initial stack-setting initialization failure, its zero-row receipt and executed sources are retained. Point counts are reused from the independently verified parent records. Ten $p = 7$ records have 13-smooth ambient order; residual cofactors are unclassified, and selected-subgroup costs are unmeasured.

The property browser is a sidecar. The earlier catalogue study and two-branch evidence remain sealed, and numerical IC ratio/census graphs retain their values: this round adds structural classification rather than a new paired timing measurement. Reproduction, source hashes, identity IDs and raw arithmetic evidence are linked in the Markdown report and validation record.

6 Sources

Sutherland: ordinary/supersingular curves and endomorphism orders.

Sutherland: isogeny volcanoes.

PARI: arithmetic and fundamental discriminants.

PARI: finite-field polynomial roots.

Parent two-branch theorem and frozen model evidence.